

Networking User Guide

Router and Switch Configuration Reference

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Chapter 1: Introduction

This guide covers the configuration and operation of common routing protocols used on enterprise network devices, including OSPF, BGP, and static routing. It is intended for network administrators configuring routers and Layer 3 switches.

Each section describes default timer values, configuration commands, and recommended practices for production deployments.

Chapter 2: VLAN Configuration

Virtual Local Area Networks (VLANs) allow a single physical switch to be logically segmented into multiple broadcast domains. VLANs are identified by a VLAN ID ranging from 1 to 4094.

Trunk ports carry traffic for multiple VLANs using 802.1Q tagging. Access ports carry traffic for a single VLAN and are typically used to connect end-user devices.

Chapter 3: OSPF Configuration

Open Shortest Path First (OSPF) is a link-state routing protocol used within a single autonomous system. Routers running OSPF exchange Hello packets to discover and maintain neighbor relationships.

3.1 Hello and Dead Intervals

The Hello interval determines how often a router sends Hello packets on an OSPF-enabled interface. The Dead interval determines how long a router waits without receiving a Hello packet before declaring a neighbor down. By default, the Dead interval is four times the Hello interval.

Network type	Default Hello interval	Default Dead interval
Broadcast (e.g. Ethernet)	10 seconds	40 seconds
Point-to-point	10 seconds	40 seconds
Non-broadcast multi-access (NBMA)	30 seconds	120 seconds

Hello and Dead intervals must match between two neighboring routers for an adjacency to form. Mismatched timers are a common cause of OSPF neighbor relationships failing to establish.

3.2 Configuring Custom Timers

To change the Hello interval on an interface, use the following command in interface configuration mode:

```
router(config-if)# ip ospf hello-interval 5
```

Lowering the Hello interval can speed up failure detection but increases control-plane traffic. Most deployments retain the default values unless fast convergence is specifically required.

Chapter 4: BGP Overview

Border Gateway Protocol (BGP) is the routing protocol used between autonomous systems on the internet. Unlike OSPF, BGP uses a path-vector approach and relies on TCP for reliable transport between peers.

BGP keepalive messages are sent every 60 seconds by default, with a hold time of 180 seconds.